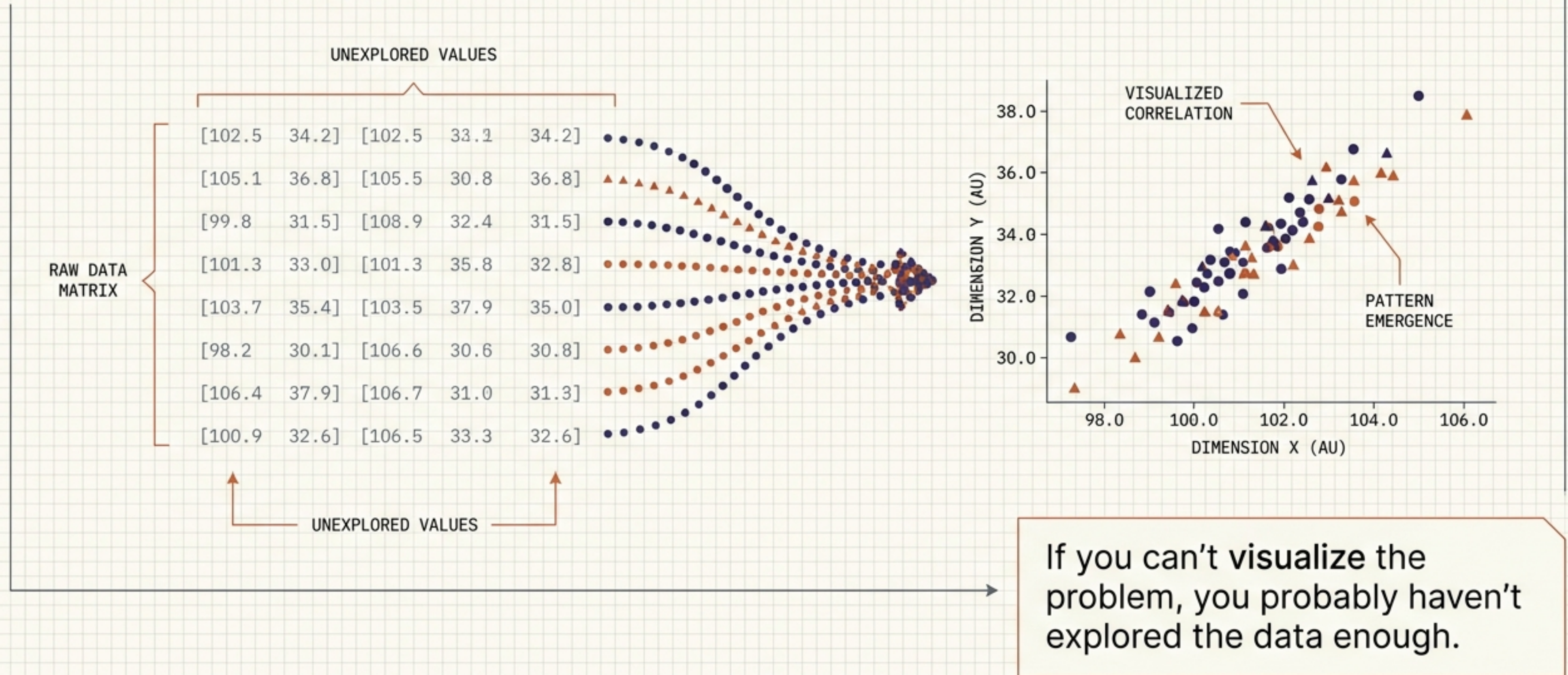


Matplotlib & Data Visualization

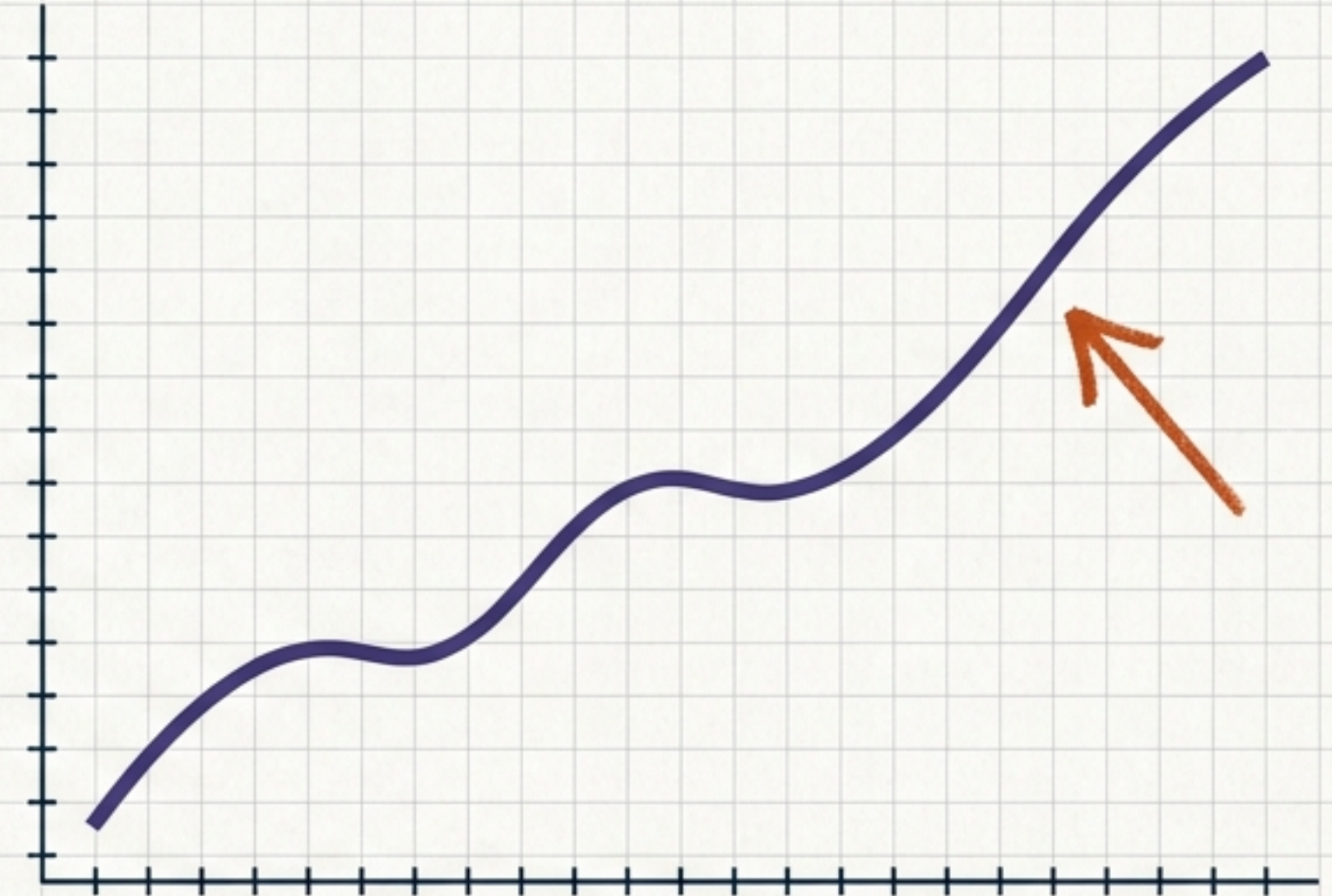
The Explorer's Guide: Learn to see your data before teaching machines.



Numbers alone hide the patterns that visual geometry reveals.

[illegible]

A dataset contains numbers.

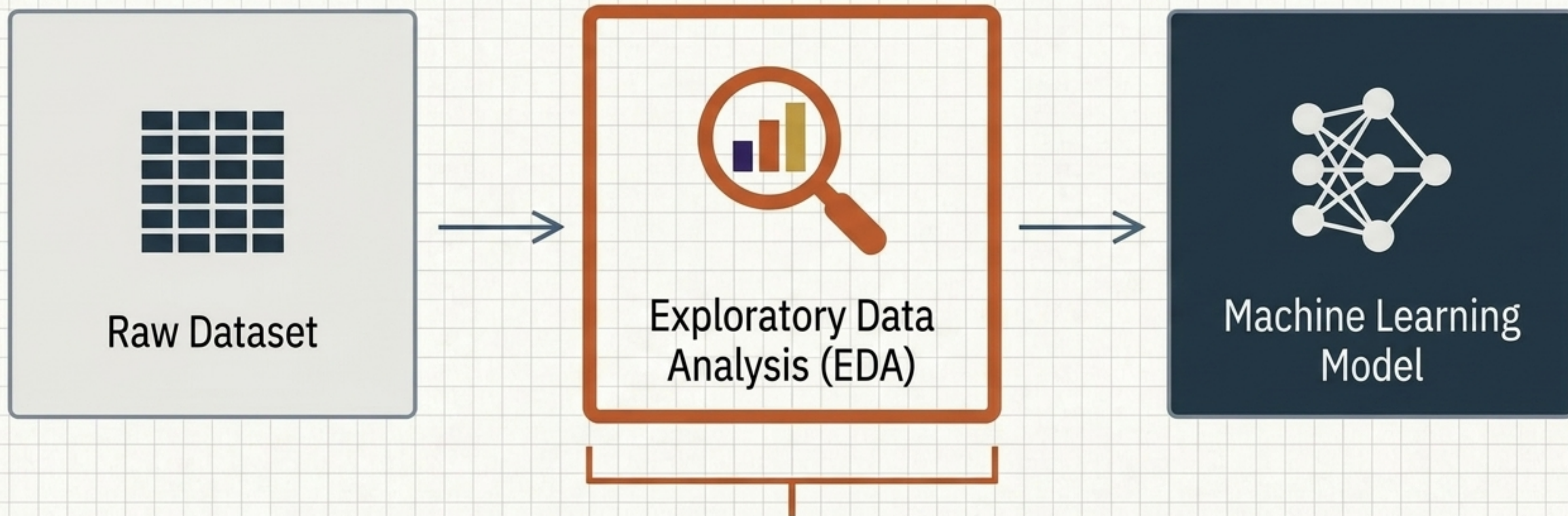


A visualization instantly reveals:
X and Y are strongly related.

Discover: Trends · Outliers · Distributions · Relationships · Patterns · Errors

Before ML: Look at the data.

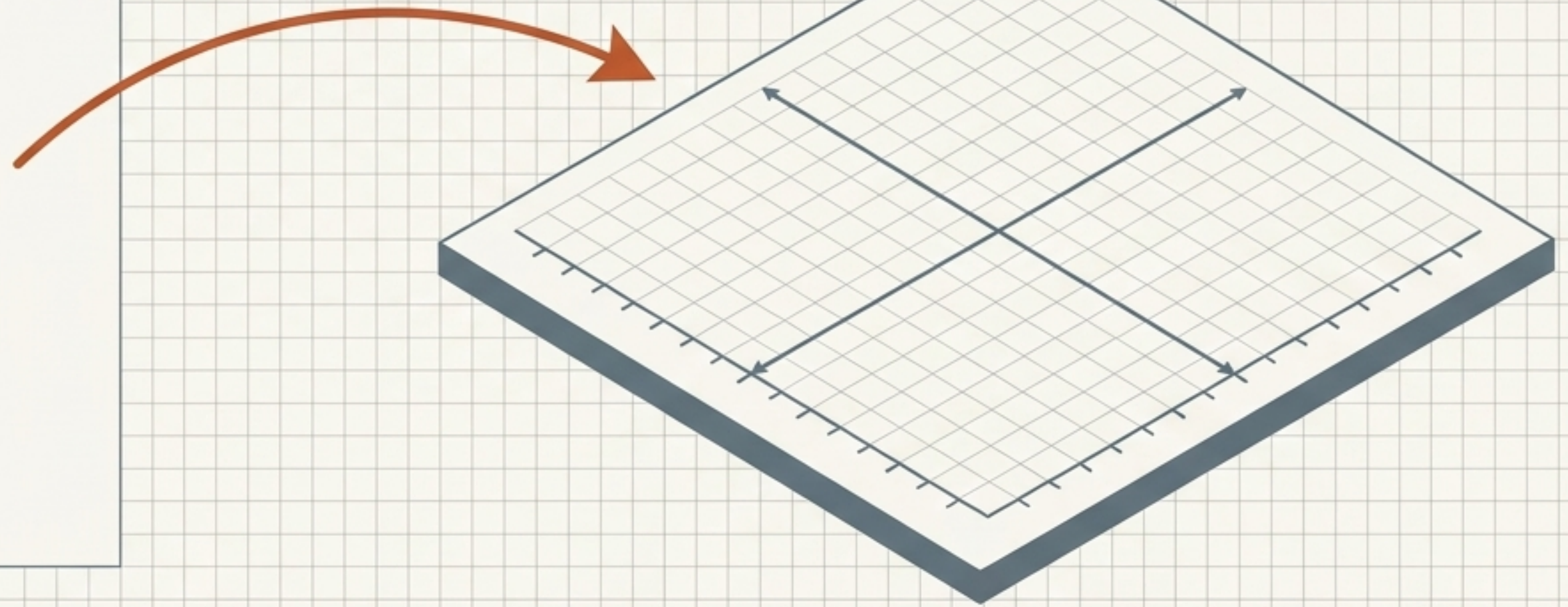
Visualization is the core reasoning tool of Exploratory Data Analysis.



Visualization is not a decorative step.
It is how we understand the dataset before modeling.
Don't blindly feed data into a black box.

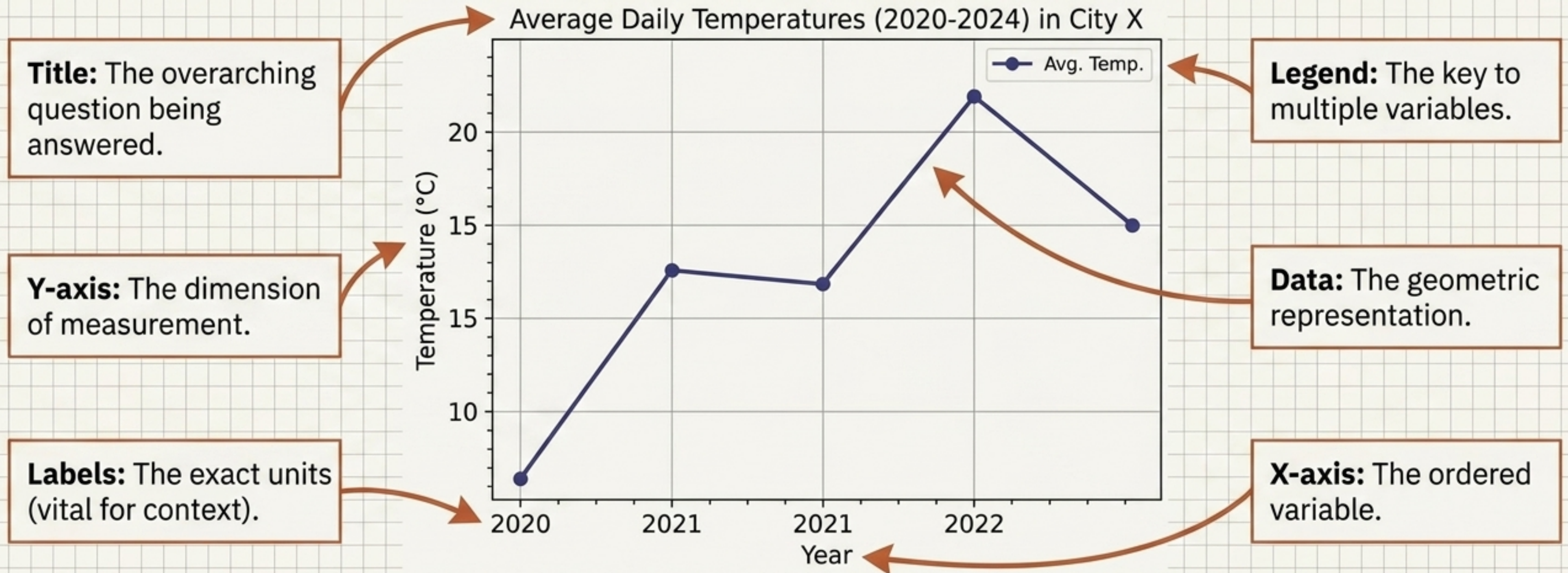
Matplotlib translates numerical arrays into visual geometry.

```
import matplotlib.pyplot as plt
```



Definition: Matplotlib is the foundational Python library used to create static, animated, and interactive visualizations. It serves as the bridge between raw code and human-readable charts.

The anatomy of a well-structured plot.



The Golden Rule: A visualization isn't just a random picture. It must communicate what the data means.

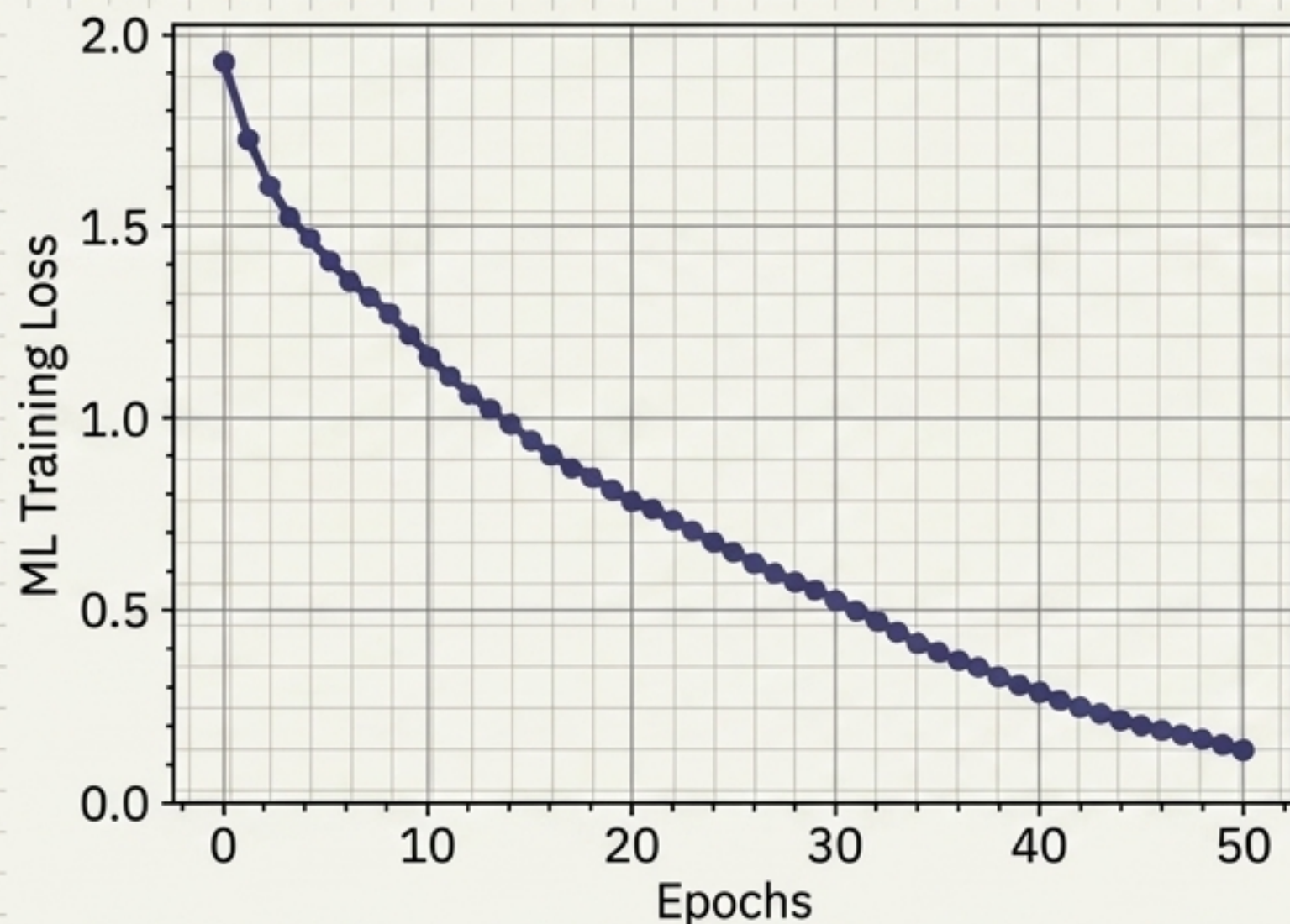
Choose your lens based on the question, not the aesthetic.

The Question	Visual Lens	Chart Type
How does something change?		Line Chart
How do categories compare?		Bar Chart
How are values distributed?		Histogram
What is the relationship between two numbers?		Scatter Plot
How can I compare multiple views?		Subplots

Rule: Don't choose a chart because it looks cool. Choose it because it answers a specific question.

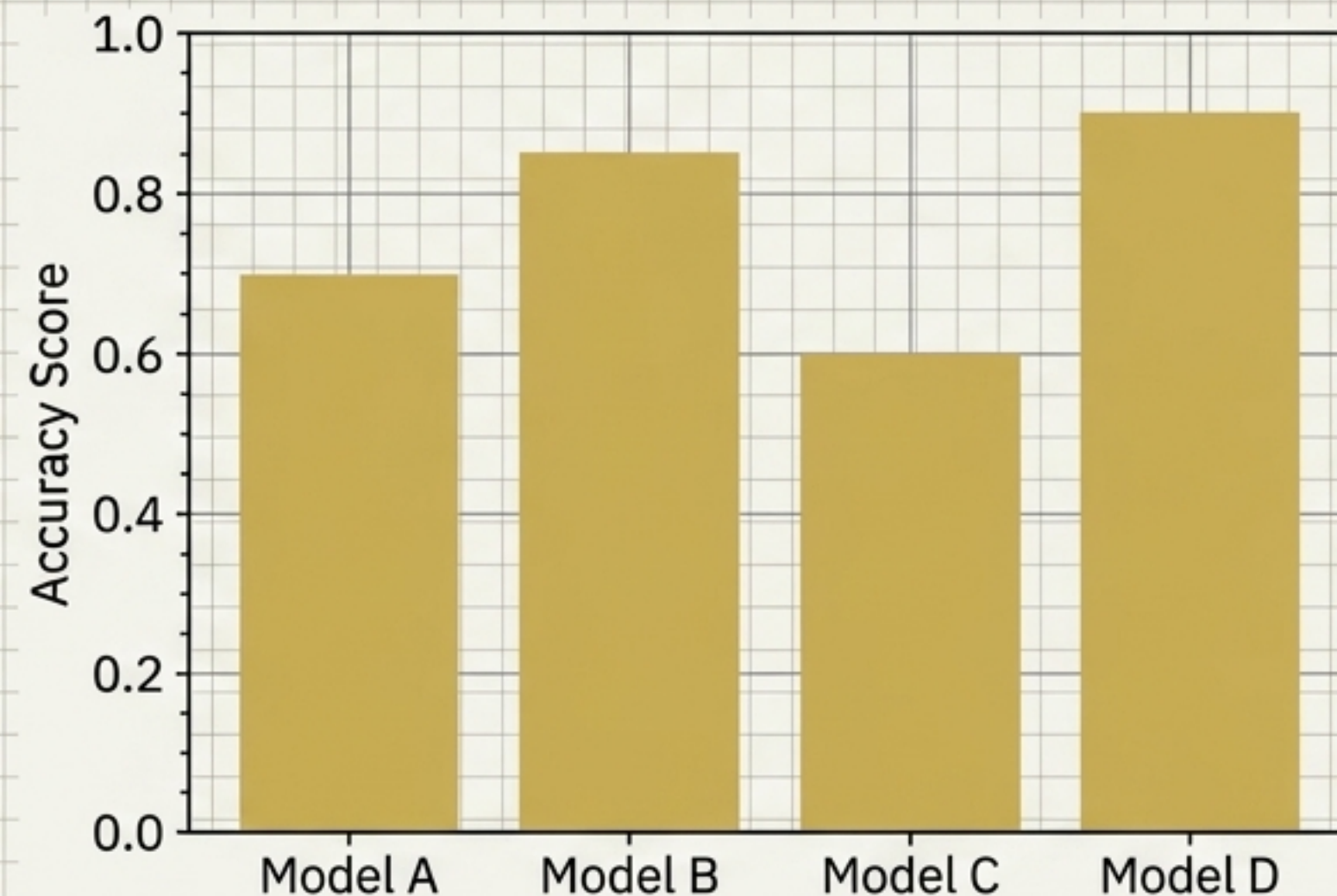
Lenses for tracking trends and comparing categories.

Line Chart



Ideal for tracking an ordered variable (often time). Shows continuous progression and trends.

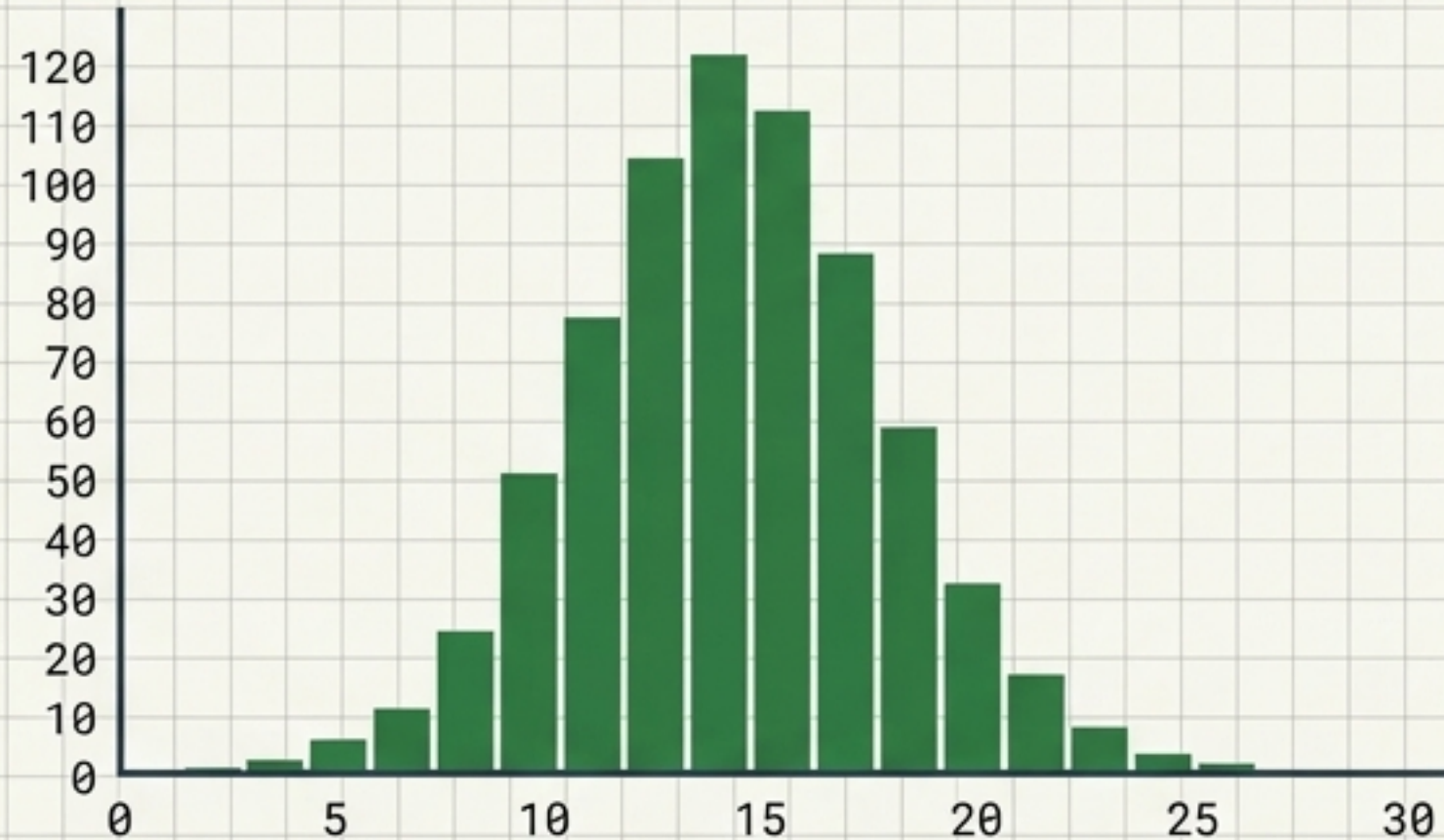
Bar Chart



Excels at category comparison, distinct counts, and contrasting discrete scores.

Lenses for mapping distributions and feature relationships.

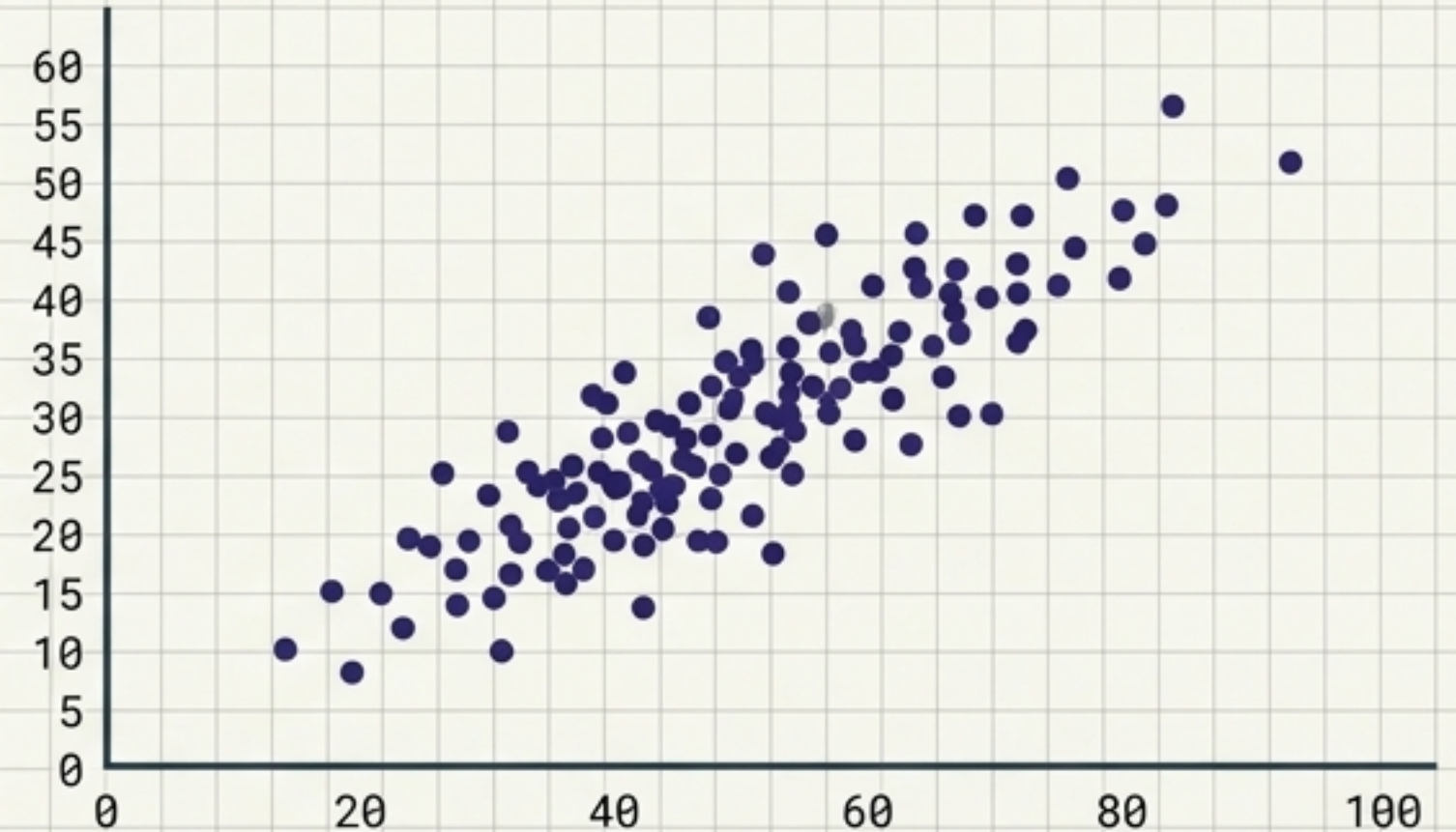
Histogram



Answers: Where are most of the values concentrated?

Used for age distributions, dataset characteristics, and error mapping.

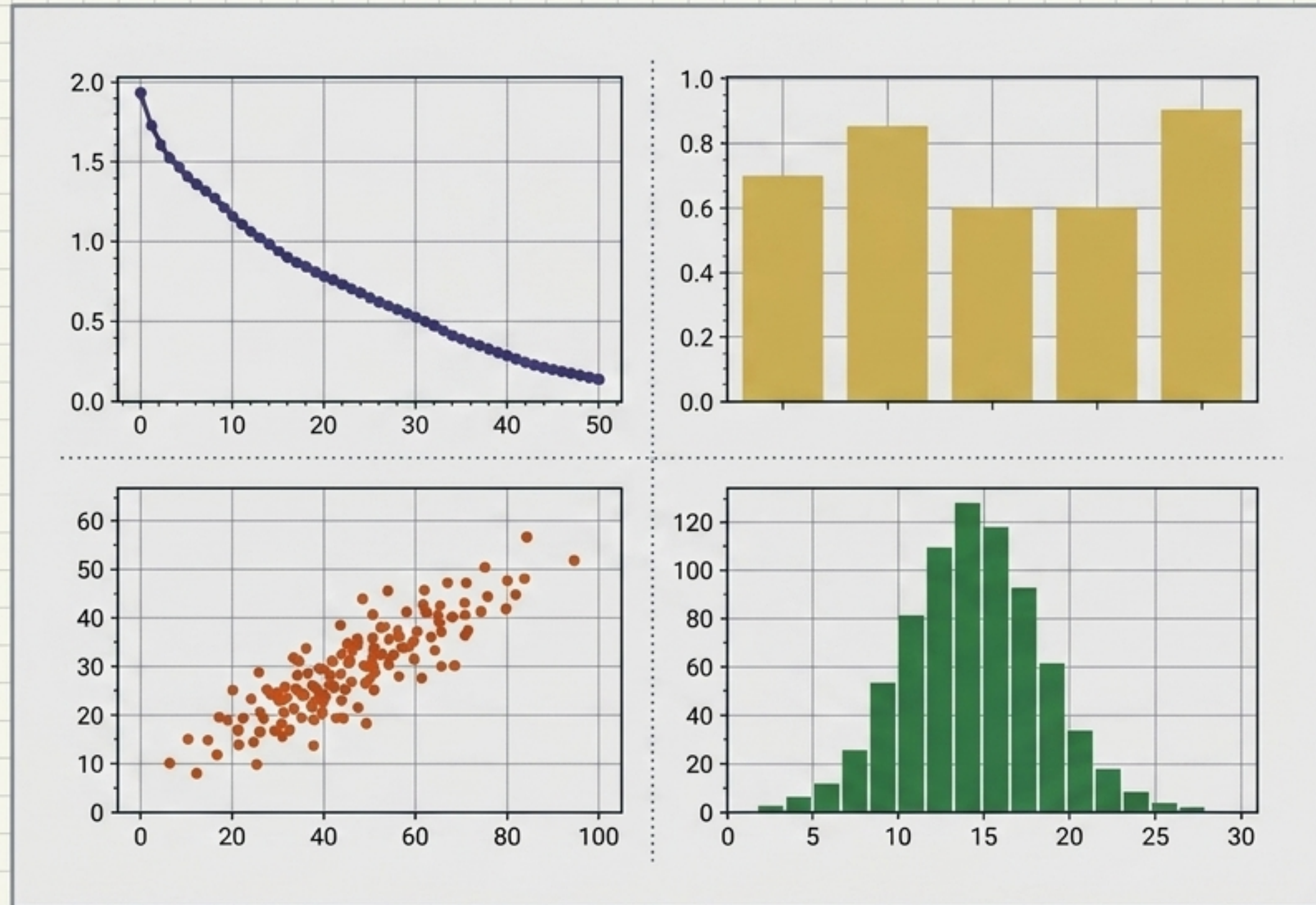
Scatter Plot



Answers: How do these two variables interact?

Used for examining relationships and correlations between distinct numerical features.

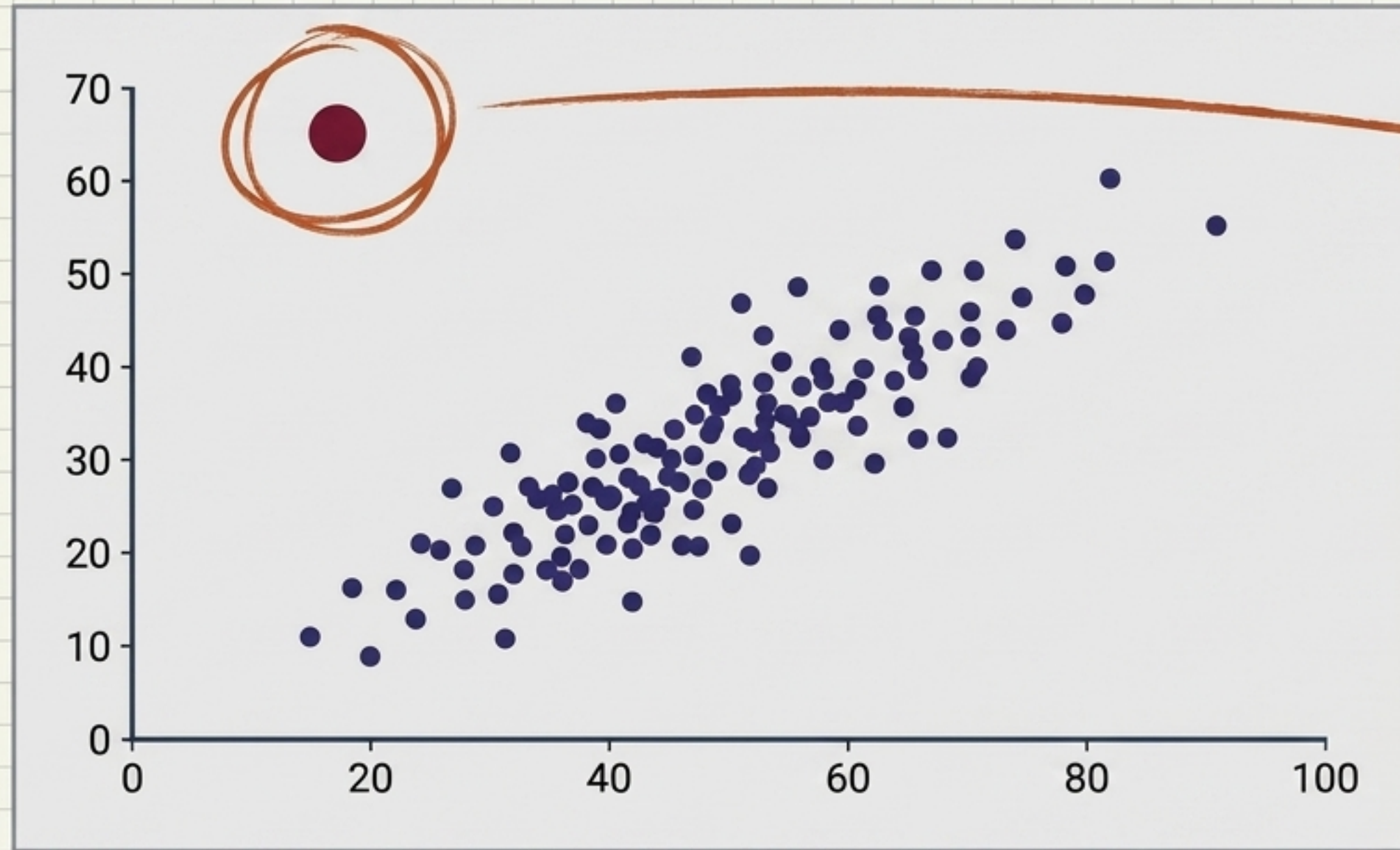
The multi-monitor view: comparing datasets with subplots.



Concept: Sometimes we need several visualizations together.

Subplots allow us to visualize multiple views of the same dataset simultaneously, side-by-side without clutter.

Visualizations highlight the anomalies; you must investigate the cause.



Investigating Outliers:

- ☐ A data-entry mistake
- ☐ A genuine unusual observation
- ☐ A unit error
- ☐ A special case

Warning on Trends: A decreasing training loss line indicates progress, but it does not automatically mean the model generalizes well. Visualization helps us investigate; it doesn't automate the decision.

A chart should explain itself without the viewer asking questions.

The Trap

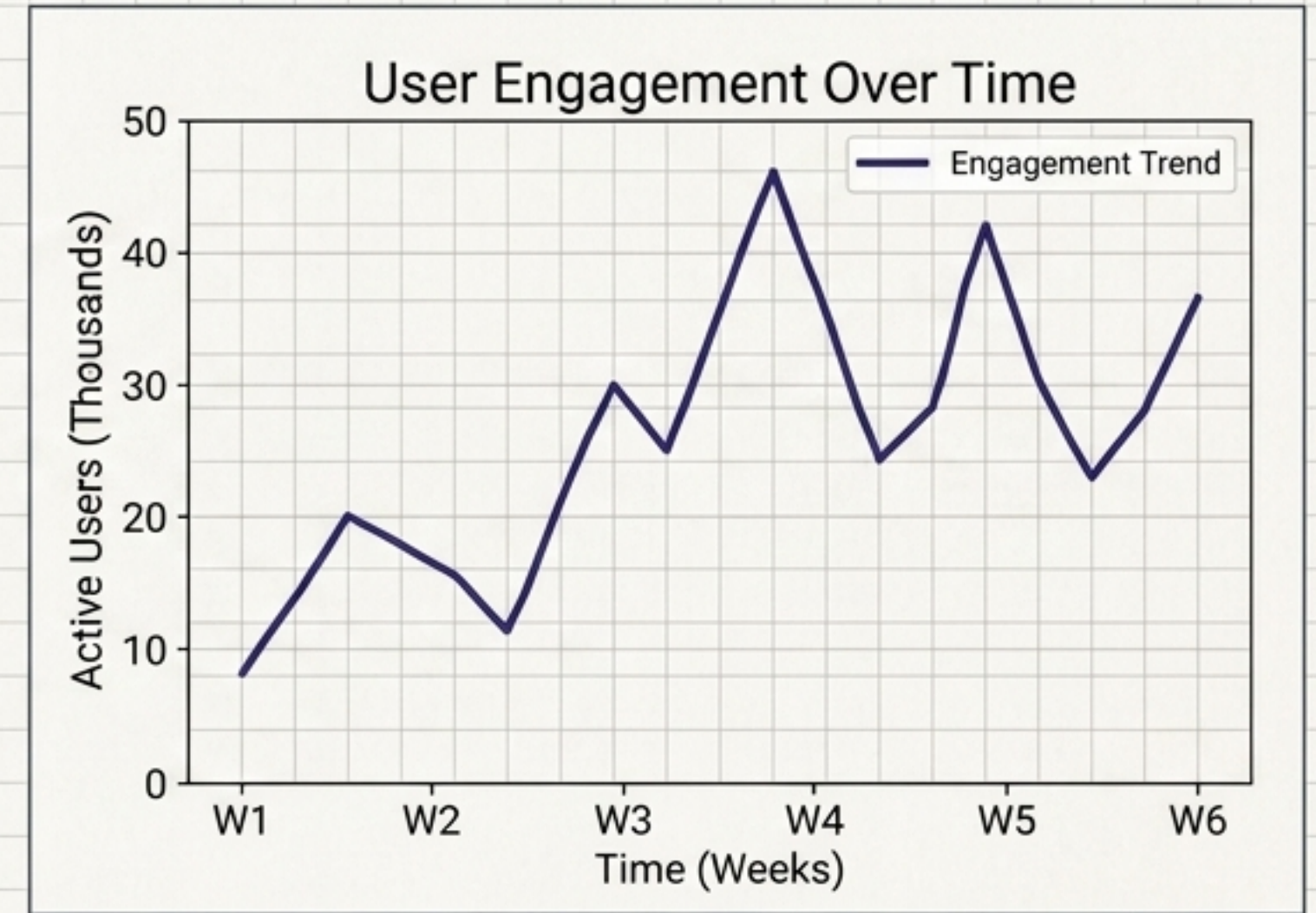
Before



What are x and y? Data without context is just ink.

The Solution

After



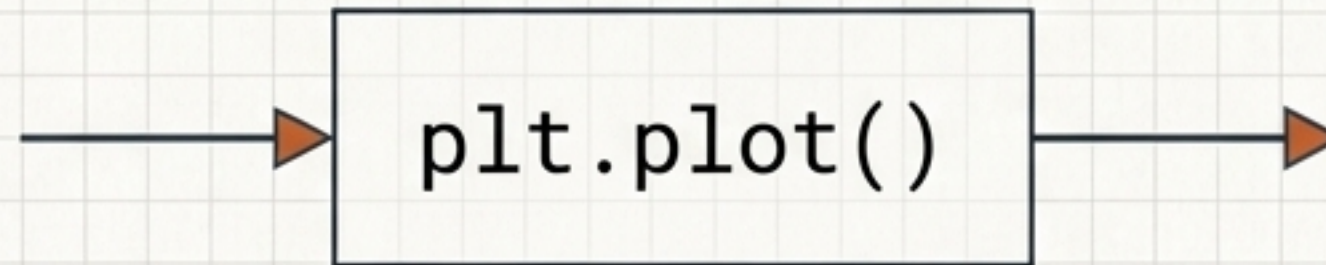
Good visualization does the cognitive work for the viewer. Meaning is instantly communicated.

Pandas wrangles the data; Matplotlib paints it.

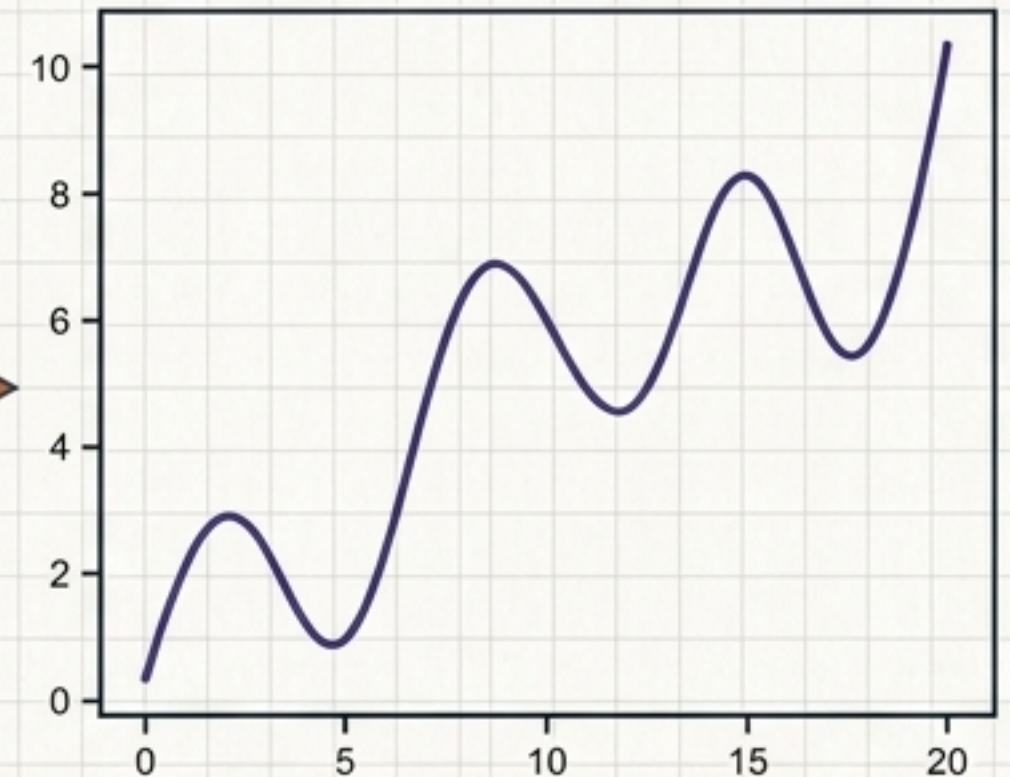
Input: Pandas DataFrame

	10.2	34.1	55.8
	12.3	45.0	60.2
	14.9	39.5	70.1

Process: Matplotlib Engine

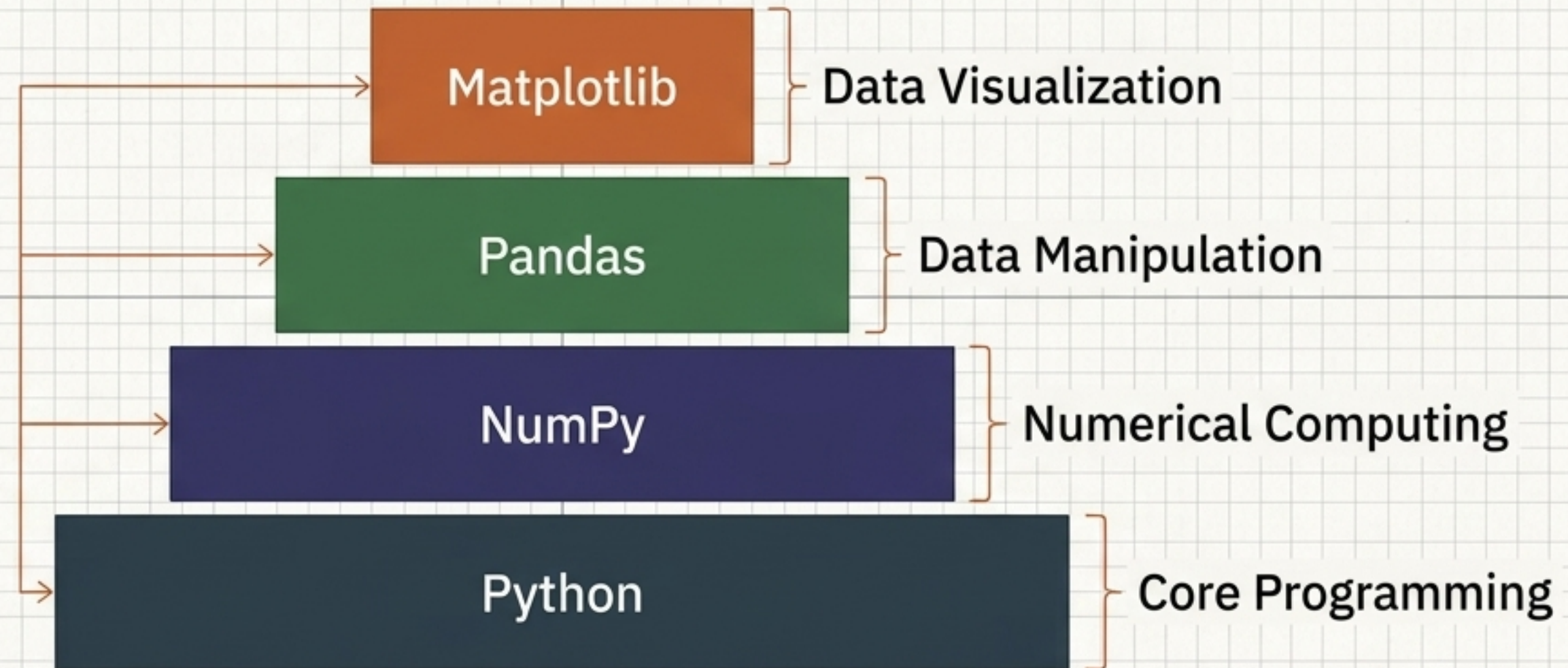


Output: Visual Geometry



Concept: Pandas and Matplotlib are designed to work together as a seamless pipeline for Exploratory Data Analysis. They are complementary halves of the EDA workflow.

The complete Python Data Stack is ready for Machine Learning.



Takeaway: We have successfully built the basic stack. Each tool has a specific role. The foundation is set for the next major chapter.

The Engineer's Manifesto: Never skip the data check.

ML Engineer: Let's just train the model.

Data: Did you check me first?

ML Engineer: I'm sure you're fine.

System Warning: Data fails silently.

ML Engineer: Maybe let's visualize this first.

The Ultimate Lesson: Data visualization isn't decoration; it's a **reasoning tool**. Understand the dataset's shape and flaws before modeling.

Mini-Practice: Patterns do not guarantee causation.

Study Hours	Scores
2 hrs	65
4 hrs	78
6 hrs	92



Diagnostic Question:
Does the dataset
appear to show a
relationship?

Challenge: Plot "Hours vs Score" as both a line plot and a scatter plot.

Final Warning: Visualization shows pattern, not necessarily causation. Do not jump straight to "Study hours cause higher scores." Rigorous testing must follow observation.